

EN.530.626: Optimal Control for Space Systems

Homework 2

Due Oct 8th 11:59PM

The code for this assignment can once again be found on GitHub at [this link](https://github.com/JHU-ACEL/ocss_hw.git) and can be downloaded by running `git clone https://github.com/JHU-ACEL/ocss_hw.git` from a terminal window. You can either clone/download the repository out or simply `git pull` if you had cloned out the repository for the last homework. Homework submissions and grading will be managed through Canvas.

Problem	Points Possible
Problem 1	20
Problem 2	20
Problem 3	15
Problem 4	5
Total	60

Introduction

This homework will focus on the following technical concepts:

1. Using Newton method to solve convex quadratic programs with linear equality and inequality constraints.
2. Synthesizing a primal-dual interior point method algorithm to satisfy the necessary conditions of optimality for such problems.

To accomplish this, the software development learning goals include,

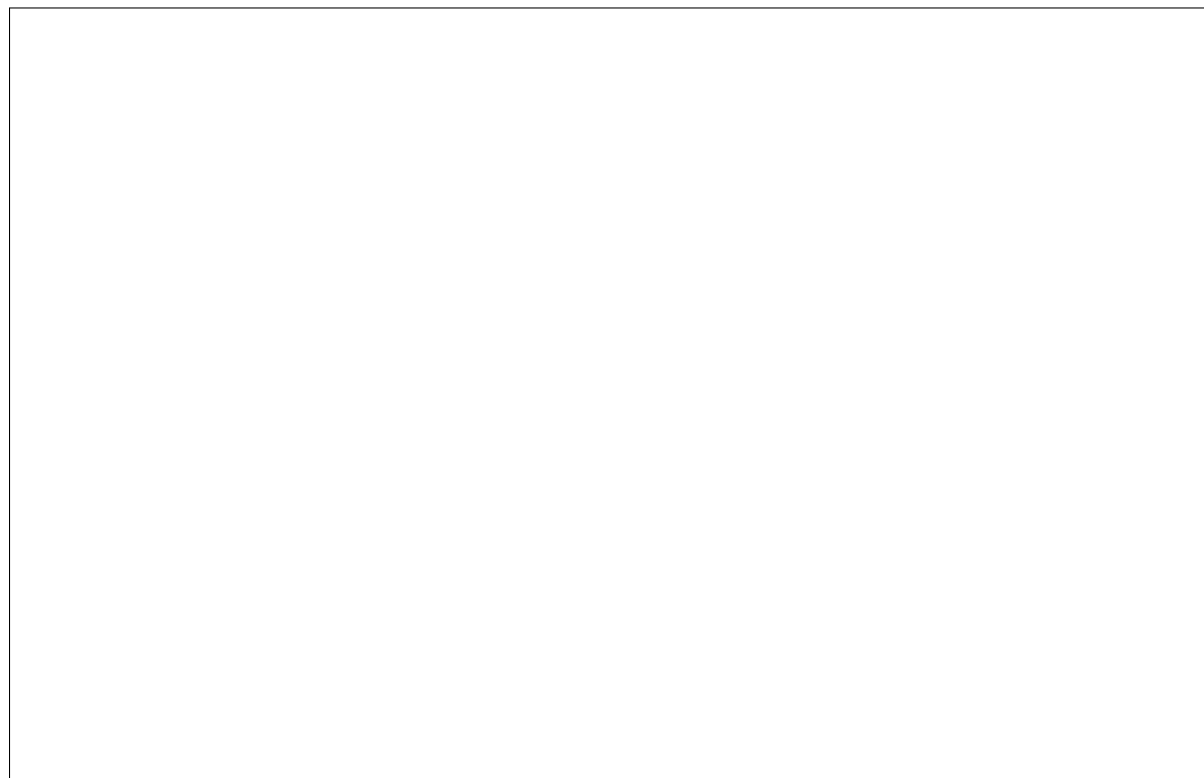
- Implementing the primal-dual interior point solver using `jax`.
- Understanding how such generic QP solvers are used to solve optimal control problems.
- Evaluating the performance of an optimal control algorithm in a robotic simulator.

1 Problem 1 [20 Points]

Part A [10 Points]

For an unconstrained optimization problem $\min_x f(x)$, where $f \in \mathcal{C}^2$ and $x \in \mathbb{R}^n$, use the definition of a local minimizer to derive the necessary conditions of optimality:

- If x^* is a local minimizer of $f(x)$, then $\nabla_x f(x^*) = 0$.
- If x^* is a local minimizer of $f(x)$, then $\nabla_{xx}^2 f(x^*) \succeq 0$ (i.e., $\nabla_{xx}^2 f(x^*)$ is positive semi-definite).



Part B [5 Points]

Find all candidates for local minima for the optimization problem

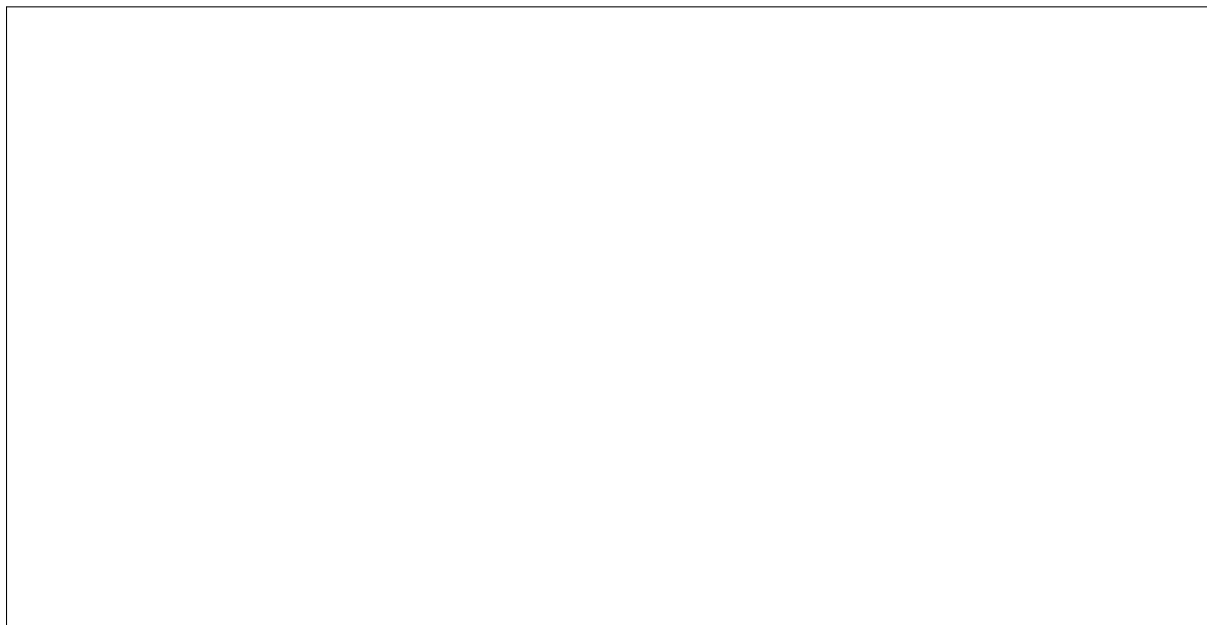
$$\begin{aligned} \min_{x_1, x_2, x_3} \quad & \frac{1}{2} (x_1^2 + x_2^2 + x_3^2) \\ \text{subject to} \quad & x_1 + x_2 + x_3 \leq -3. \end{aligned}$$

**Part C [5 Points]**

Consider the problem,

$$\begin{aligned} \min_{x_1, x_2} \quad & -2x_1 - x_2 \\ \text{subject to:} \quad & -x_1^3 + 9x_1^2 - 24x_1 + x_2 \leq 0, \\ & -x_2 + 20 \leq 0, \\ & x_1 - 7 \leq 0. \end{aligned}$$

Does the point $(x_1, x_2) = (2, 20)$ satisfy the KKT conditions for this problem?



2 Problem 2: A Vanilla Convex QP Solver [20 Points]

In this problem, we will be implementing the quadratic program solver from [1] to solve a problem of the form,

$$\begin{aligned} & \underset{x}{\text{minimize}} \quad \frac{1}{2}x^T Px + p^T x \\ & \text{subject to} \quad Ax = b \\ & \quad \quad \quad Gx \leq h \end{aligned}$$

where $x \in \mathbb{R}^n$, $P \in \mathbf{S}_+^n$, and $p \in \mathbb{R}^n$. The linear equality constraints are given by $A \in \mathbb{R}^{n_{\text{eq}} \times n}$ and $b \in \mathbb{R}^{n_{\text{eq}}}$. The linear inequality constraints are $G \in \mathbb{R}^{n_{\text{ineq}} \times n}$ and $h \in \mathbb{R}^{n_{\text{ineq}}}$.

Part A [5 Points]

By using the Lagrange multiplier theorem, show how to derive the initial tuple $(x^{(0)}, s^{(0)}, z^{(0)}, y^{(0)})$ by deriving the necessary conditions of optimality for,

$$\begin{aligned} & \underset{x}{\text{minimize}} \quad \frac{1}{2}x^T Px + q^T x + \frac{1}{2}\|s\|_2^2 \\ & \text{subject to:} \quad Ax = b, \\ & \quad \quad \quad Gx + s = h. \end{aligned}$$

Can this initial iterate be directly used? What is the size of each column of this matrix?

Part B [15 Points]

3 Problem 3: Optimal Control as a Quadratic Program [15 Points]

Next, take the “vanilla” quadratic program solver from Problem 1 and use it to solve a simple optimal control problem for a 2D spacecraft:

$$\begin{aligned}
 & \underset{q_{0:N}, u_{0:N-1}}{\text{minimize}} && q_N^T Q_f q_N + \sum_{k=0}^{N-1} u_k^T R u_k + q_k^T Q q_k \\
 & \text{subject to} && q_{k+1} = A_{\text{sc}} q_k + B_{\text{sc}} u_k, \quad k = 0, \dots, N-1 \\
 & && q_0 = q_{\text{init}}, \\
 & && q_{\min} \leq q_k \leq q_{\max}, \quad k = 1, \dots, N, \\
 & && u_{\min} \leq u_k \leq u_{\max}, \quad k = 0, \dots, N-1.
 \end{aligned}$$

Our spacecraft state will be modeled as $q_t = (p_t, v_t) \in \mathbb{R}^{n_x}$, where p_t and v_t are the position and velocity, respectively. The acceleration applied to the system $u_t \in \mathbb{R}^{n_u}$. As this is a 2D problem, $n_x = 4$ and $n_u = 2$. We will model the dynamics as a “double integrator”, wherein the system dynamics are linear time-invariant (LTI):

$$q_{t+1} = A_{\text{sc}} q_t + B_{\text{sc}} u_t = \begin{pmatrix} I_{2 \times 2} & \Delta h I_{2 \times 2} \\ 0_{2 \times 2} & I_{2 \times 2} \end{pmatrix} q_t + \begin{pmatrix} 0.5 \Delta h^2 I_{2 \times 2} \\ \Delta h I_{2 \times 2} \end{pmatrix} u_t, \quad (1)$$

where Δh is the discretization time.

Part A [5 Points]

Our quadratic program will solve for the full state and control trajectory over some horizon N such that our decision variables will be $x = (q_0, u_0, q_1, u_1, \dots, q_N)$ ¹ and we will enforce these dynamics as linear equality constraints $Ax = b$ in our quadratic program.

The quadratic program interface we wrote in Problem 2 requires a single matrix expression $Ax = b$ — using the A_{sc} and B_{sc} matrices from (1), write out the structure of A and b and indicate their dimensionality in terms of n_x and n_u .

¹The autograder will assume that the state q_t and control u_t variables are ordered exactly this way.

A large, empty rectangular box with a thin black border, intended for the student's answer to the question in Part A.

Part B [10 Points]

A rectangular box with a thin black border, intended for the student's answer to the question in Part B. It is smaller than the box in Part A.

Problem 4: mujoco Simulation [5 Points]

Now, take optimal control problem solver from Problem 3, and the “vanilla” quadratic program solver from Problem 2, and use it to make a wrapper class to be used inside a mujoco simulator. The simulation is of the Mars Ingenuity Helicopter ². We will ignore the specific details of helicopter control here and only focus on the simulating a simplified 2D double integrator problem. The goal of the simulation is to approach the origin from a specified initial point on the xy plane.

- In the file, `mpc.py`, complete the implementation of the class methods.
- These methods will be called inside the file `mhs_sim_test.py`, and will be used inside the mujoco simulation.
- To test the simulation, run `start_viewer.sh` in the terminal and open `localhost:6080` on your browser. There, click on `vnc.html`, and connect to viewer to view a display of the mujoco simulation on your browser.

Deliverables:

- Add screenshots of the simulation to show the helicopter reaching the goal position, which is the origin. A marker has been added to assist with this.

²<https://github.com/nasa-jpl/m2020-urdf-models>

Submission Instruction

- Download the marimo notebook for each problem and join them together into a single PDF or HTML file named `hw2.pdf`.
- Compress the `hw2` folder containing your python files. Name this file “`hw2.zip`”.
- Upload both the PDF/HTML file and the zip folder into the canvas assignment “Homework 2”.

References

- [1] J. Mattingley and S. Boyd, “CVXGEN: A code generator for embedded convex optimization,” Optimization and Engineering, vol. 12, no. 1, pp. 1–27.